

# Celltac $\alpha$ +

Automated Hematology and ESR Analyzer

MEK-1305

NIHON KOHDEN

Celltac  $\alpha$ +



**3-part differential  
hematology analyzer  
with ESR**

*Fighting Disease with Electronics*

NIHON KOHDEN

# Integration

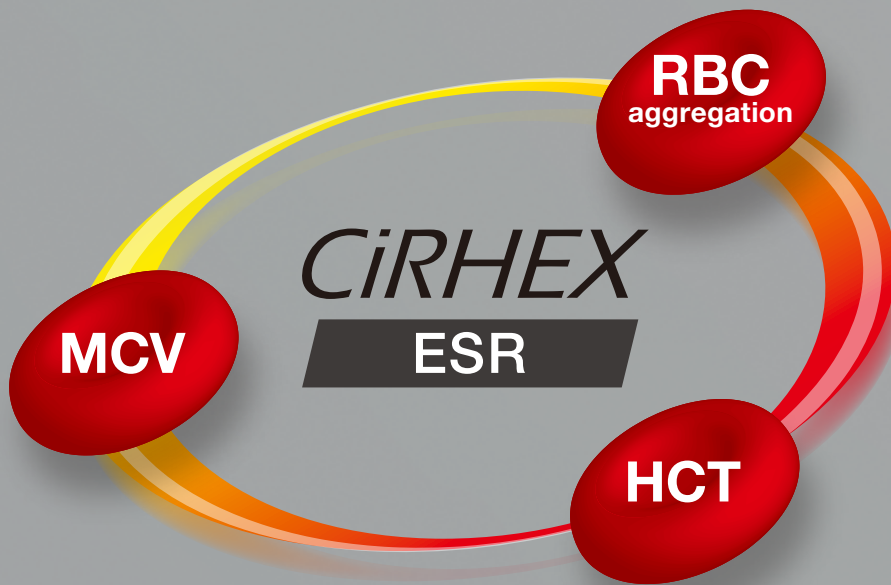
Transforming the possibility of IVD solutions

Infectious disease is spreading all over the world and becoming a serious problem. For example, 10 million people developed Tuberculosis (TB) which is one of the most serious infectious diseases, and TB caused an estimated 1.6 million deaths in 2017 according to data from the WHO.

What are your laboratory challenges?

What is required for a better clinical outcome?

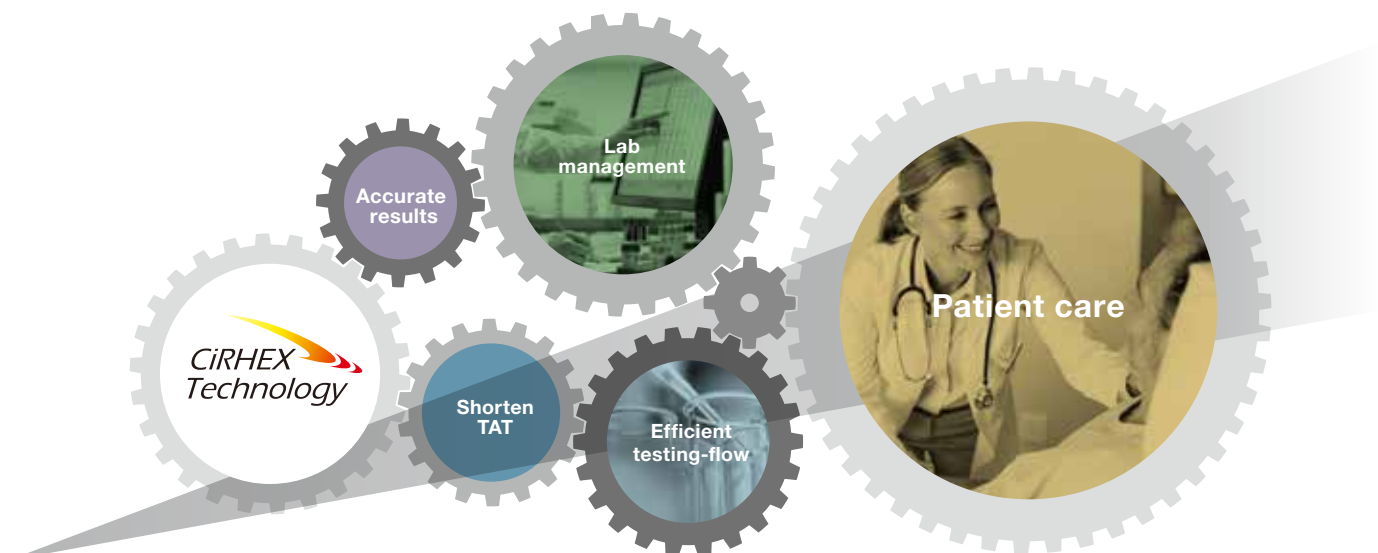
Celltac  $\alpha$ +, equipped with ESR, can help to achieve a better clinical outcome.



CiRHEX (Cell counter integrated rheometric excellence) technology can provide ESR result highly correlated with Westergren method by using HCT value and MCV value from CBC measurement and also RBC aggregation phenomenon.



# Nihon Kohden's unique CiRHEX Technology helps you to achieve a better clinical outcome



## Result displayed in 2 minutes by single EDTA tube for both CBC and ESR

A single EDTA tube can be used for both CBC and ESR measurement on our Celltac α+, and you will be able to get CBC results on the screen in 1 minute, and an ESR result in 2 minutes, with a single aspiration. This leads to reducing your workload, avoiding the risk of infection and providing a quick report to the patient.



## Only 80 µL sample volume required for both CBC and ESR

Unlike the traditional methods for ESR testing, our Celltac α+ requires only 80 µL of blood sample for both CBC and ESR measurement. Not only does this small blood collection volume improve your laboratory workflow, it also improves the patient experience.



## No additional reagents, no additional cost

As it is based on conventional hematology analyzers, diluent, detergent and hemolysing reagent are used for CBC and 3 part differential measurement on Celltac α+, but no additional reagents are required for ESR measurement.



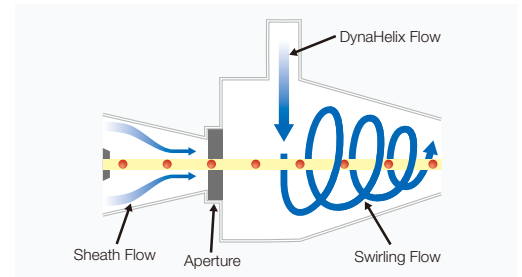
# Innovation

Maximizes laboratory productivity

## Quality hematology testing

DynaHelix Flow technology perfectly aligns RBC and PLT cells for high impedance counting precision using an advanced hydrodynamic-focused sheath flow before passing through the aperture. In addition, the DynaHelix Flow totally prevents the risk of coincidence or re-entry of counted blood cells into the aperture, using the unique DynaHelix Flow stream.

This newly-developed advanced DynaHelix Flow Technology greatly improves counting precision and accuracy.



## Integrated QC program



- QC program for ESR is available
- The same QC material can be used for CBC, 3 part diff and ESR
- QC lot management up to 25
- Assay value registration using a handy barcode reader (standard accessory)
- Automated judgement function (pass or fail)
- QC management by assay value, average value or Westgard multirule
- QC graph display and printout (optional)
- Automated calculation of statistical information such as average and SD

## Reagent management



Standard accessory, barcode reader



Celltac α+ reagent management system helps easier reagent bottle management with a unique barcode labeled on each reagent. Through this system and use of genuine Nihon Kohden reagents, testing quality is always maintained at a high level.

## Thirty-two measuring parameters including ESR and other research parameters

Traditional CBC parameters, WBC 3 part differential parameters, Mentzer Index and RDWI, which are considered to be useful for Thalassemia screening. ESR and other parameters which are related to ESR are available on Celltac α+.

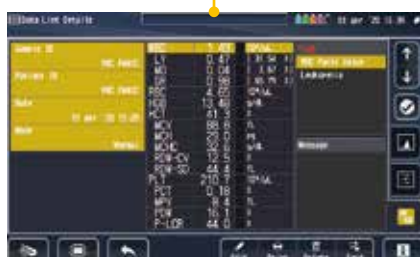
# Operational excellence

Smart  
ColoRerun Assist

Smart ColoRerun Assist helps to visually understand the reasons of re-measurement, by showing color-coded messages. This unique user-oriented function greatly improves workflow efficiency and maximizes productivity for faster test reports and clinical decision making.

## YELLOW

A panic value (far outside the normal range) needs to be reported to a doctor immediately



## ORANGE

Possibly incorrect data due to problems caused by the state of the blood sample or the measuring procedure



## RED

Possibly incorrect data due to a technical problem with the instrument or measuring procedure



## Seamless information transfer

### Seamless Information Transfer



Celltac α+ supports seamless data transfer\* to laboratory information systems through the LAN port or RS-232C port.

\* ASTM protocol is available

### Sister product

## Celltac $\alpha$ MEK-1301/1302

### Innovative hematology platform offering

- High quality CBC measurement based on DynaHelix Flow technology
- Smart ColoRerun Assist visually showing the reasons of re-measurement
- 23 measuring parameters including WBC 3 part differential
- Up to 60 samples/hour throughput (open mode)
- Complete QC program for laboratory accreditation requirements



MEK-1301  
(open mode only)



MEK-1302  
(open and closed mode)

## Key Specifications

■ **Number of measuring parameters: 32**  
WBC, LY%, MO%, GR%, LY#, MO#, GR#, RBC, HGB, HCT, MCV, MCH, MCHC, RDW-CV, RDW-SD, PLT, PCT, MPV, PDW, P-LCR, P-LCC\*, Mentzer Index\*, RDWI\*, NLR\*, ESR, ESR HCT Corr.\* ESR TEMP Corr.\*, SA\*, AMP\*, AI\*, MIN\*, t1/2\*

\* Research parameters

\* Research parameters

■ **Measuring mode:** Open mode

### ■ Throughput

CBC + WBC 3 part differential: Approx. 60 samples/h  
CBC + WBC 3 part differential + ESR: Approx. 20 samples/h

CBC + WBC 3 part differential + ESR: Approx. 20 samples/h

### ■ Measuring time

CBC + WBC 3 part differential: Approx. 45 s/ sample\*

CBC + WBC 3 part differential + ESR: Approx. 95 s/ sample\*

\* from measurement start to data display

\* from measurement start to data display

### ■ Sample volume

Normal mode: CBC + WBC 3 part differential 20μL  
                  : CBC + WBC 3 part differential + ESR 80μL

Predilution mode: CBC 10 or 20μL

Capillary mode: CBC 20μL

Capillary mode: CBC 20µL

### ■ **Measuring method**

WBC, RBC and PLT count: Electric impedance method  
(DynaHelix Flow technology)

HGB: Colorimetric method

HCT: Calculated from RBC histogram

WBC differential: Calculated from WBC histogram

ESR: Calculated from sylectogram, HCT and MCV  
(CiRHEX technology)

ESR: Calculated from syllectogram, HCT and MCV  
(CiRHEX technology)

### ■ Measuring range

- WBC: 0.00 - 99.99 x 10<sup>3</sup>/μL, 0.00 - 299.90 x 10<sup>3</sup>/μL (High dilution mode)
- RBC: 0.00 - 9.99 x 10<sup>6</sup>/μL
- HGB: 0.00 - 29.90 g/dL
- HCT: 0.0 - 99.9%
- MCV: 20.0 - 199.0 fL
- MCH: 10.0 - 50.0 pg
- MCHC: 10.0 - 50.0 pg
- PLT: 0.0 - 1490.0 x 10<sup>3</sup>/μL
- ESR: 0 - 200 mm

ESR: 0 - 200 mm

- **Data storage capacity:** 50,000 data including histograms in the memory of the analyzer

## Reproducibility and Linearity

### ■ Reproducibility

- WBC: 2.0% or less (WBC:  $4.00 \times 10^3/\mu\text{L}$  or more)
- RBC: 1.5% or less (RBC:  $4.00 \times 10^6/\mu\text{L}$  or more)
- HGB: 1.5% or less
- HCT: 1.5% or less
- MCV: 1.0% or less
- MCH: 2.0% or less
- MCHC: 2.0% or less
- PLT: 4.0% or less (PLT:  $100.0 \times 10^3/\mu\text{L}$  or more)
- ESR: 10.0% or less, or SD 1.5 mm or less

ESR: 10.0% or less, or SD 1.5 mm or less

### ■ Linearity

WBC: Within  $\pm 3.00\%$  or  $\pm 0.30 \times 10^3/\mu\text{L}$  (WBC: 0.20 to  $99.9 \times 10^3/\mu\text{L}$ )  
RBC: Within  $\pm 3.00\%$  or  $\pm 0.08 \times 10^6/\mu\text{L}$  (RBC: 0.02 to  $8.00 \times 10^6/\mu\text{L}$ )  
HGB: Within  $\pm 1.50\%$  or  $\pm 0.20 \text{ g/dL}$  (HGB: 0.10 to 25.0 g/dL)  
HCT: Within  $\pm 3.0\%$  or  $\pm 1.0\%$  (HCT: 20.0 to 60.0%)  
PLT: Within  $\pm 10.0\%$  or  $\pm 20.0 \times 10^3/\mu\text{L}$  (PLT: 10.0 to  $1490.0 \times 10^3/\mu\text{L}$ )  
(specifications above apply to normal mode)

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## Physical Specifications

■ **Dimensions:** 230 W x 450 D x 428 H mm

■ **Weight:** 21 kg

- **Line voltage:** 100 V to 240 V

- **Line frequency:** 50 or 60 Hz

- **Power input:** 150 VA

■ **External output:** LAN x 1, USB x 2, RS-232C x 3

## Environmental Conditions

- **Operating temperature:** 15 to 30°C

- **Operating humidity:** 30 to 85%

- **Operating atmospheric pressure:** 700 to 1060 hPa

## Reagent

- **Diluent:** Isotonac 3 or Isotonac 4

■ **Hemolysing reagent:** Hemolynac 310

■ **Detergent:** Cleanac 710, Cleanac 3